Sales Data Analysis System

Project No : 4

Team Contribution

Poojitha - Application integration and dashboard

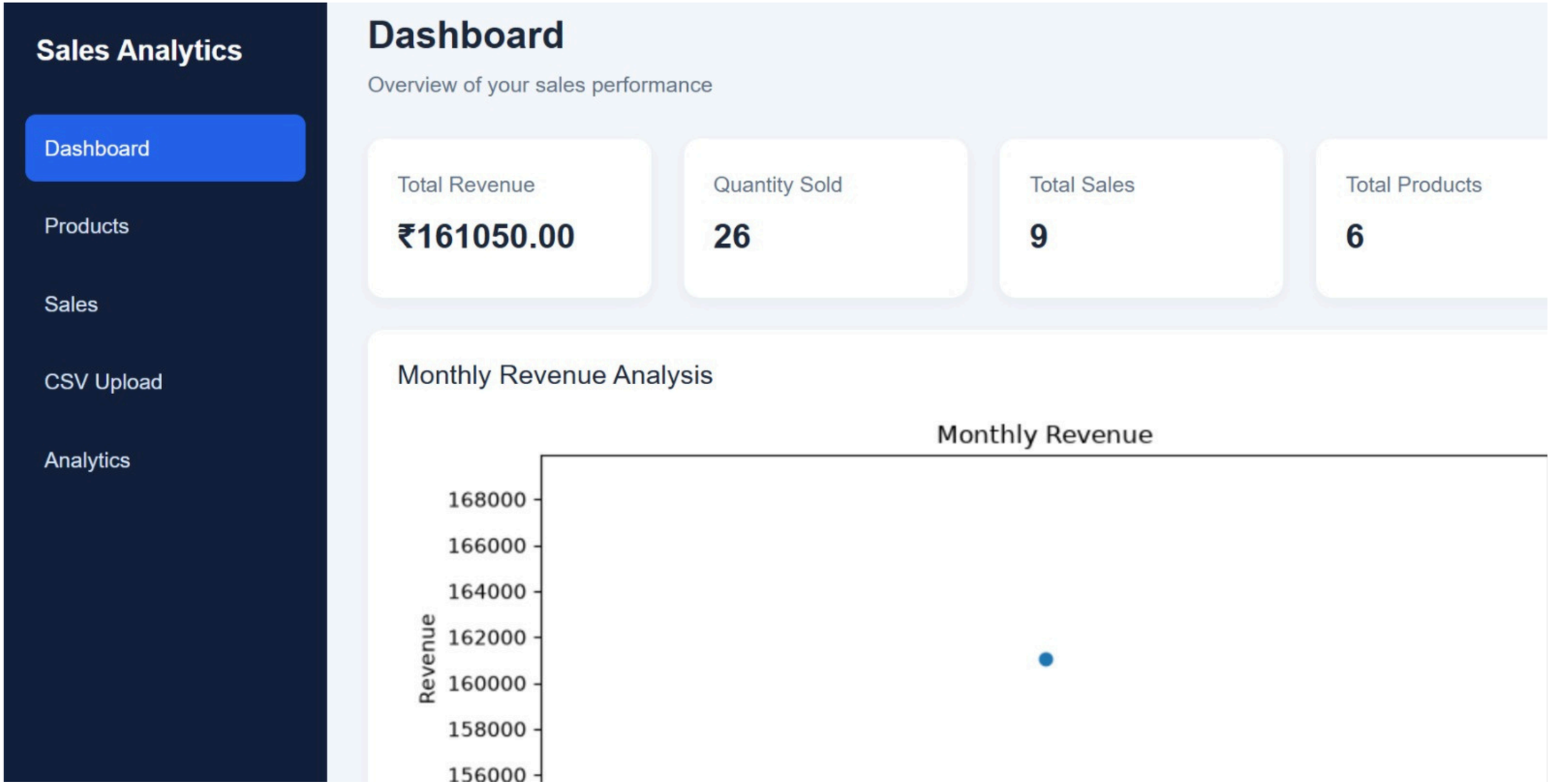
Harshini - Product management

Greeshma - Sales records and database

Swetha - CSV upload and validation

Premsai - Analytics, charts and reports

Visual Representation



Implementation and Working

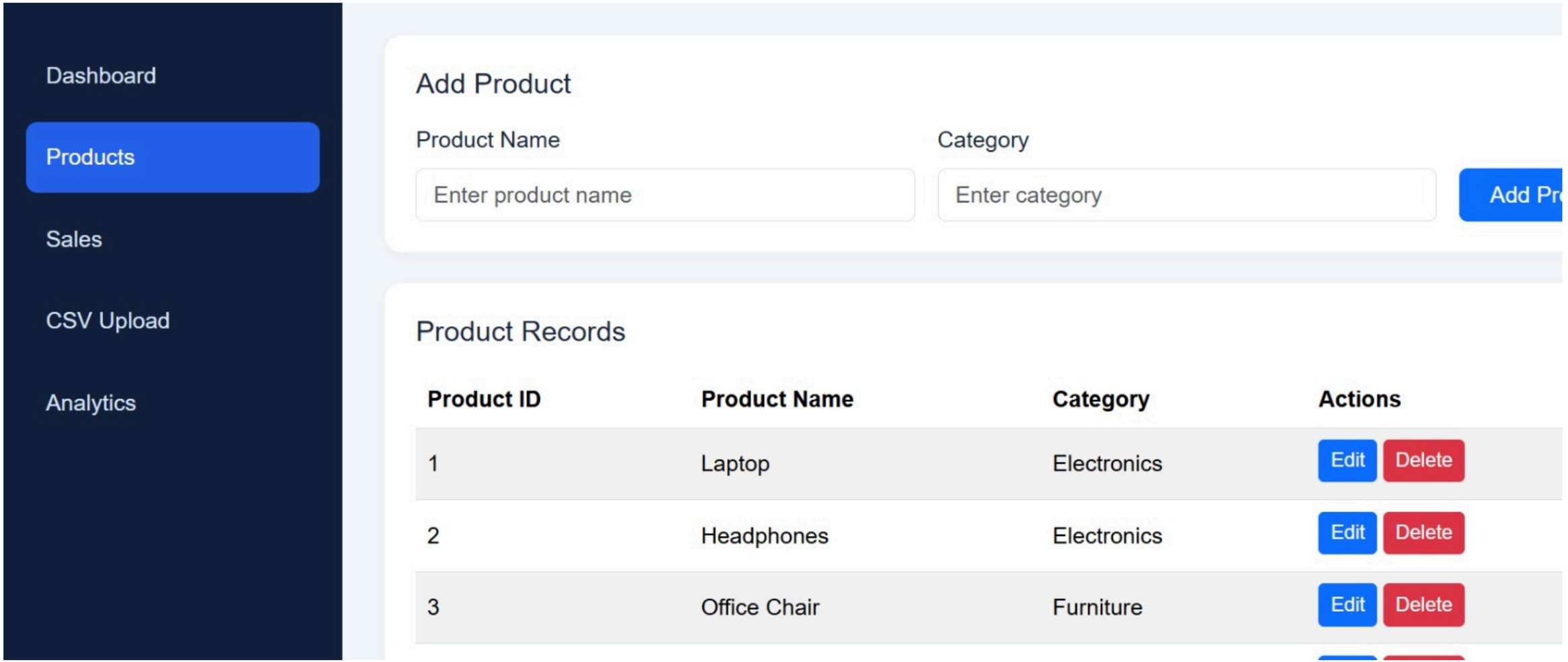
Technologies Used

- Python
- Flask
- MySQL
- HTML
- CSS
- JavaScript

Key Responsibilities

- Integrated all project modules using Flask.
- Connected the application to the MySQL database.
- Displayed sales summaries on the dashboard.
- Coordinated module integration and tested the application.

Visual Representation



Implementation and Working

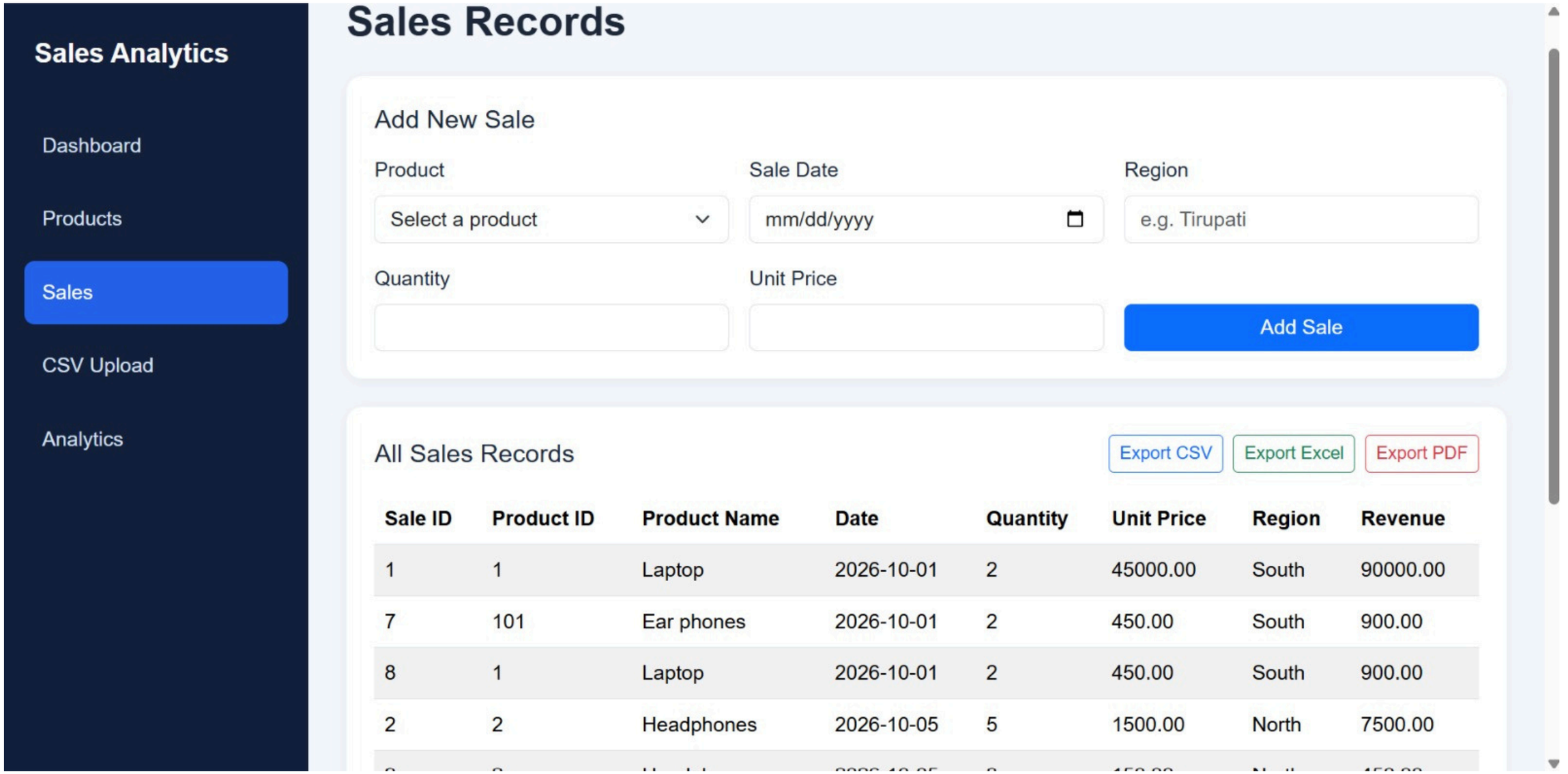
Technologies Used

- Python
- Flask
- MySQL
- HTML
- CSS
- JavaScript

Key Responsibilities

- Stores product names and categories.
- Saves product information in the MySQL database.
- Allows users to add and edit products.
- Validates product inputs and manages deletion rules.

Visual Representation



Implementation and Working

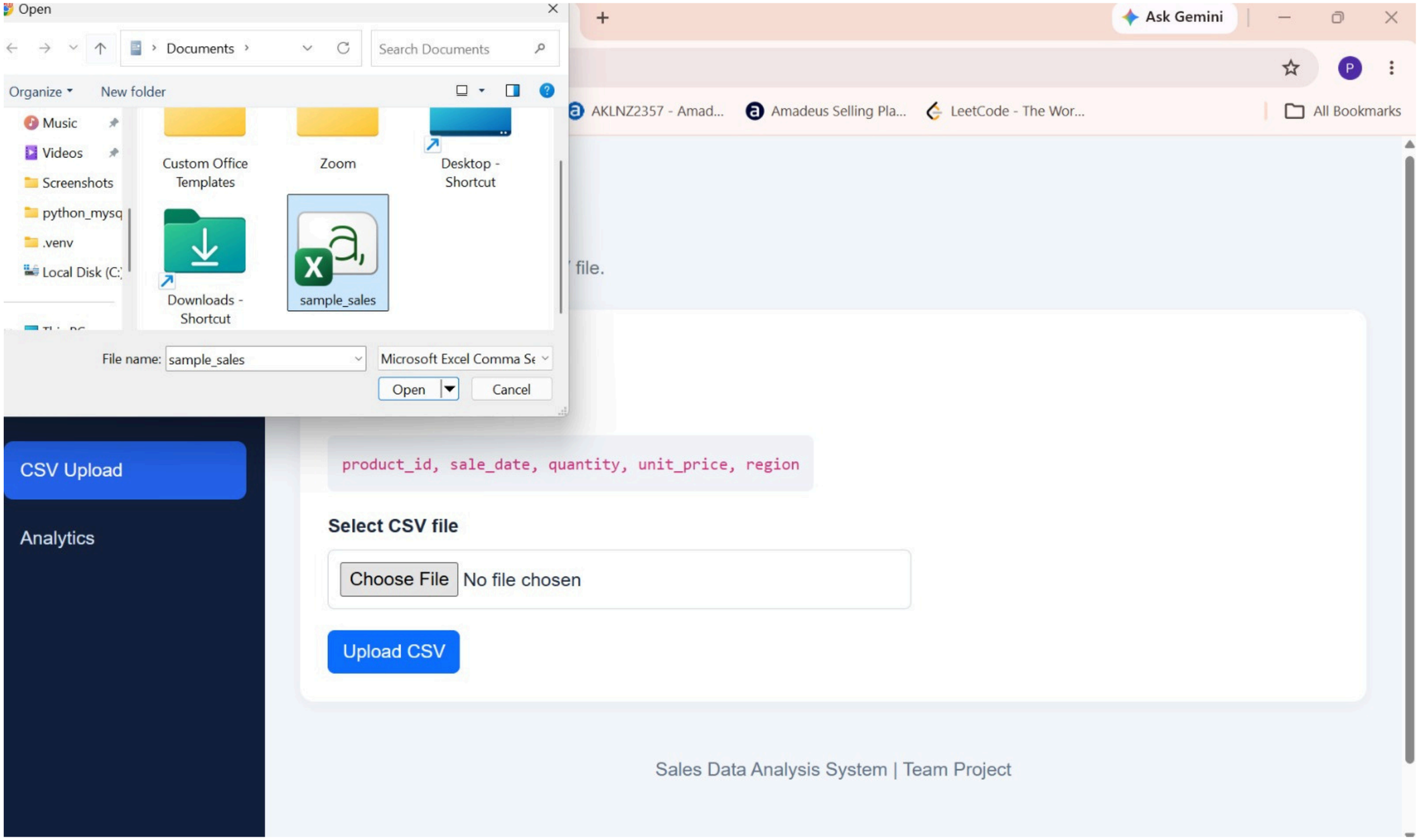
Technologies Used

- Python
- Flask
- MySQL
- HTML
- CSS
- JavaScript

Key Responsibilities

- Stores product ID, sale date, quantity, unit price and region.
- Uses product IDs to connect sales to products.
- Calculates revenue as quantity × unit price.
- Supports adding, editing and deleting sales records.

Visual Representation



Implementation and Working

Technologies Used

- Python
- Flask
- Pandas
- MySQL

Key Responsibilities

- Reads structured CSV files using Pandas.
- Checks required columns and data values.
- Checks whether product IDs exist.
- Imports valid data into MySQL.
- Reports errors when input data is invalid.

Visual Representation

Sales Analytics

Dashboard

Products

Sales

CSV Upload

Analytics

Detailed Sales Analytics

Analyze revenue, quantities, products, categories, regions, and monthly performance.

Filter by Date

Start Date

mm/dd/yyyy

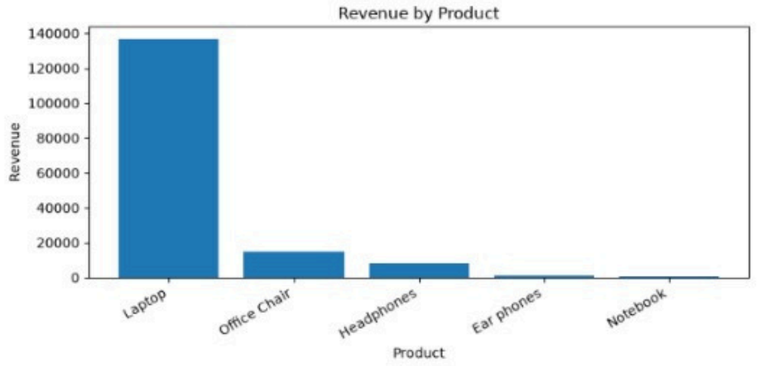
End Date

mm/dd/yyyy

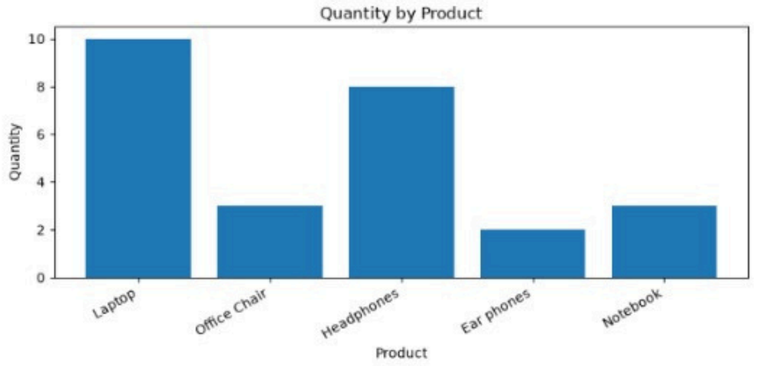
Apply Filter

Reset

Revenue by Product



Quantity Sold by Product



Implementation and Working

Technologies Used

- Python
- Pandas
- Matplotlib
- NumPy
- CSV
- Flask
- PDF / Excel Libraries

Key Responsibilities

- Uses Pandas to group and summarize sales data.
- Uses Matplotlib to display charts.
- Compares revenue and quantity across products.
- Shows monthly and regional trends.
- Supports report exports, where implemented.


Thank You